Regression Output Analysis

Oct 25, 2026 · @Bryan

Logistic Regression: Output

```
Logistic Regression
Response: low
Predictors: smoke, ht, ui
Rows used: 189

                 Generalized Linear Model Regression Results
==============================================================================
=
Dep. Variable:                    low   No. Observations:
189
Model:                            GLM   Df Residuals:
185
Model Family:                Binomial   Df Model:
3
Link Function:                  Logit   Scale:
1.0000
Method:                          IRLS   Log-Likelihood:
-110.00
Date:                Sun, 25 Oct 2026   Deviance:
220.01
Time:                        15:48:49   Pearson chi2:
188.
No. Iterations:                     4   Pseudo R-squ. (CS):
0.07467
Covariance Type:            nonrobust
==============================================================================
=
                 coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
-
const         -1.3551      0.241     -5.611      0.000      -1.828
-0.882
smoke          0.6831      0.329      2.075      0.038       0.038
1.329
```

```
ht             1.4174      0.624      2.273      0.023       0.195
2.640
ui             1.0384      0.428      2.426      0.015       0.200
1.877
==============================================================================
=

Odds ratios (exp of coefficients):
const    0.257925
smoke    1.980064
ht       4.126470
ui       2.824678

Confusion matrix (threshold = 0.5):
Predicted    0   1
Actual
0          119  11
1           45  14

Accuracy: 0.7037
```

Logistic Regression: Answers

1. **Accuracy:** 0.7037, about 70.4%. That is (119 + 14) / 189 = 133 correct predictions out of 189.
2. **Lowest precision:** 0.56 for class 1 (low = 1). Precision is true positives divided by all predicted positives, so 14 / (14 + 11) = 14 / 25 = 0.56. Class 0 precision is higher at 119 / (119 + 45) = 0.726.
3. **Support of at least 30 in each category:** Yes. Actual class 0 has 130 observations (119 + 11) and actual class 1 has 59 (45 + 14). Both are above 30.
4. **Is the model good, based on pseudo-R^2:** No. The pseudo-R^2 (Cox & Snell) is 0.0747, so the predictors (smoke, ht, ui) explain only about 7% of the variation in the outcome. The 70% accuracy is mostly the model favoring the majority class: it finds only 14 of the 59 actual class 1 cases (recall about 0.24). The three predictors are statistically significant ($p < 0.05$), but the model is weak overall.

Count-Variable Regression: Output

```
Poisson (Count-Variable) Regression
Response: mhlth_crudeprev
Predictors: binge_crudeprev, checkup_crudeprev, csmoking_crudeprev,
depression_crudeprev, obesity_crudeprev, sleep_crudeprev, selfcare_crudeprev,
indeplive_crudeprev, loneliness_crudeprev, foodinsecu_crudeprev,
housinsecu_crudeprev
Rows used: 2299

                 Generalized Linear Model Regression Results
==============================================================================
=
Dep. Variable:        mhlth_crudeprev   No. Observations:
2299
Model:                            GLM   Df Residuals:
2287
Model Family:                 Poisson   Df Model:
11
Link Function:                    Log   Scale:
1.0000
Method:                          IRLS   Log-Likelihood:
-5397.6
Date:                Sun, 25 Oct 2026   Deviance:
39.316
Time:                        15:59:30   Pearson chi2:
39.2
No. Iterations:                     4   Pseudo R-squ. (CS):
0.1925
Covariance Type:            nonrobust
==============================================================================
===========
                            coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
-----------
const                     1.7032      0.172      9.888      0.000       1.366
2.041
binge_crudeprev           0.0032      0.003      0.954      0.340      -0.003
0.010
checkup_crudeprev        -0.0004      0.002     -0.192      0.848      -0.004
0.004
csmoking_crudeprev        0.0043      0.004      1.100      0.271      -0.003
```

```
0.012
depression_crudeprev     0.0137      0.002      5.800      0.000       0.009
0.018
obesity_crudeprev        0.0001      0.002      0.076      0.940      -0.003
0.004
sleep_crudeprev          0.0024      0.002      1.059      0.289      -0.002
0.007
selfcare_crudeprev      -0.0608      0.022     -2.708      0.007      -0.105
-0.017
indeplive_crudeprev      0.0522      0.014      3.759      0.000       0.025
0.079
loneliness_crudeprev     0.0113      0.003      3.717      0.000       0.005
0.017
foodinsecu_crudeprev    -0.0083      0.007     -1.247      0.212      -0.021
0.005
housinsecu_crudeprev     0.0150      0.008      1.786      0.074      -0.001
0.032
==============================================================================
===========

Incidence rate ratios (exp of coefficients):
const                   5.491486
binge_crudeprev         1.003177
checkup_crudeprev       0.999605
csmoking_crudeprev      1.004280
depression_crudeprev    1.013747
obesity_crudeprev       1.000134
sleep_crudeprev         1.002388
selfcare_crudeprev      0.941040
indeplive_crudeprev     1.053541
loneliness_crudeprev    1.011331
foodinsecu_crudeprev    0.991775
housinsecu_crudeprev    1.015145

Variance Inflation Factors (VIF):
            variable       VIF
     binge_crudeprev  2.528842
   checkup_crudeprev  1.750444
  csmoking_crudeprev  6.934794
depression_crudeprev  2.216376
   obesity_crudeprev  2.573017
     sleep_crudeprev  3.209334
  selfcare_crudeprev 30.514413
```

```
 indeplive_crudeprev 30.953309
loneliness_crudeprev  3.432565
foodinsecu_crudeprev 62.830787
housinsecu_crudeprev 46.839954
```

Count-Variable Regression: Answers

1. **Significant predictors (P>|z| < 0.05):** four of them.
   - depression_crudeprev (coef 0.0137, p = 0.000)
   - selfcare_crudeprev (coef -0.0608, p = 0.007)
   - indeplive_crudeprev (coef 0.0522, p = 0.000)
   - loneliness_crudeprev (coef 0.0113, p = 0.000)
2. **Strongest predictor by coefficient:** selfcare_crudeprev, with the largest absolute coefficient (-0.0608). Its incidence rate ratio is 0.941, so each 1-point increase in self-care prevalence is associated with about a 5.9% decrease in the expected mental health prevalence. indeplive_crudeprev is next (coef 0.0522, IRR 1.054). If "most significant" means smallest p-value instead, depression, indeplive and loneliness all show p = 0.000, and selfcare actually has the largest p-value of the four.
3. **Overinflated VIF (> 10) among significant predictors:** Yes, two of them. selfcare_crudeprev has a VIF of 30.51 and indeplive_crudeprev has a VIF of 30.95, which points to multicollinearity, so their coefficients may be unstable. depression_crudeprev (VIF 2.22) and loneliness_crudeprev (VIF 3.43) are fine. The other high-VIF variables, foodinsecu (62.83) and housinsecu (46.84), are not significant.
4. **Variability captured:** The pseudo-R^2 (Cox & Snell) is 0.1925, so the model captures about 19.3% of the variability. That is modest, and pseudo-R^2 is only an approximation of the R^2 in ordinary linear regression.